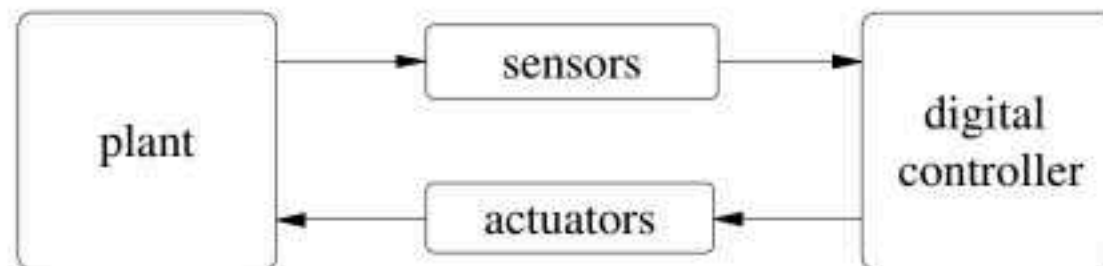


Outline

- Autonomous Systems and Their Correctness Verification (Introduction)
- Formal Methods Practice: Recently developed Tool framework
- Formal Methods Theory(as Tools of the Trade)
 - Model Checking
 - Static Analysis
 - Abstract Interpretation
- Open Challenges

Employed in diverse areas such as ...



- Enable automation: simple ON/OFF and/or sophisticated control
- New incarnations: CPS, IoT, Mechatronics, etc.
- Exciting incarnations: Autonomous systems, robotic systems, drones, UAVs, AUVs, etc.
- Uniqueness:
 - Real-time
 - Non-terminating
 - Infinite states (cont. dynamics)
 - Hybrid
- Most importantly, majority of them are SAFETY CRITICAL



Safety criticality mandates established correctness

Safety critical systems

Their failure might have catastrophic consequences, such as:

- Loss of life and
 - Huge financial implications
-
- We need to have high confidence in their correctness
 - Evidence-for that they behave correctly (as expected)
 - Solution: Formal Methods

Autonomous Systems: What it means?

Autonomous Control Systems

Control systems with a power and ability for self-governance in the performance of control functions

- setting the goal/s and
 - planning the course of actions to achieve the goal/s.
-
- Must perform well under significant uncertainties of the plant/environment
 - For extended periods of time!!!!
 - involves controllers implementing High-level decision making techniques for reasoning under UC and to act accordingly
 - requires intelligent autonomous controllers

Autonomous Systems include ...



Autonomous Under-Water Vehicles



100

- Autonomous systems rely on intelligent inference capabilities to be able to take appropriate actions even in unforeseen circumstances.
 - It is typically very difficult to assess the reliability of autonomy software.
-
- For this reason, UAV/Drones are restricted to “Segregated” airspace
 - Lack of confidence in their correctness
 - Certification authorities would soon be commissioned
 - Ex: RTCA (also EUROCAE) defined standards such as DO-178 178-B and 178-C
 - Correctness, not absolute but as per specification!!!!

Formal Verification (FV)

- applies mathematics to solve the system (HW/SW) correctness problem

Definition (System correctness problem)

R: requirement (what) D: design (how)

D conforms with R

- involves: formal specification and proof methods

Formal Verification (FV)

- applies mathematics to solve the system (HW/SW) correctness problem

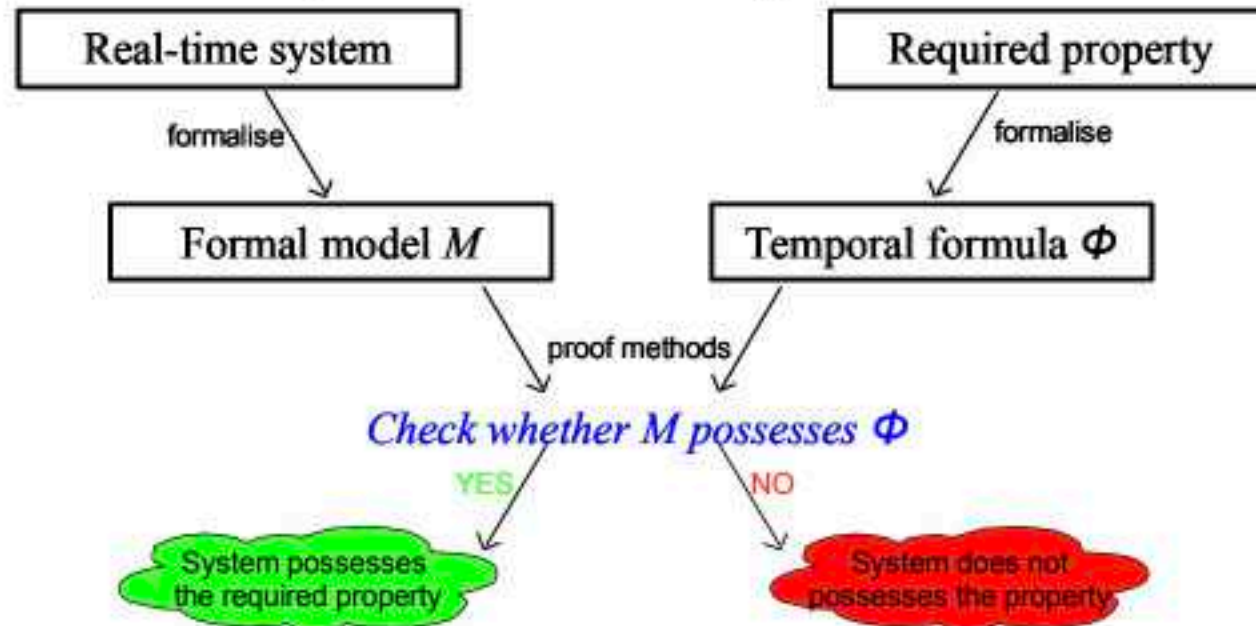
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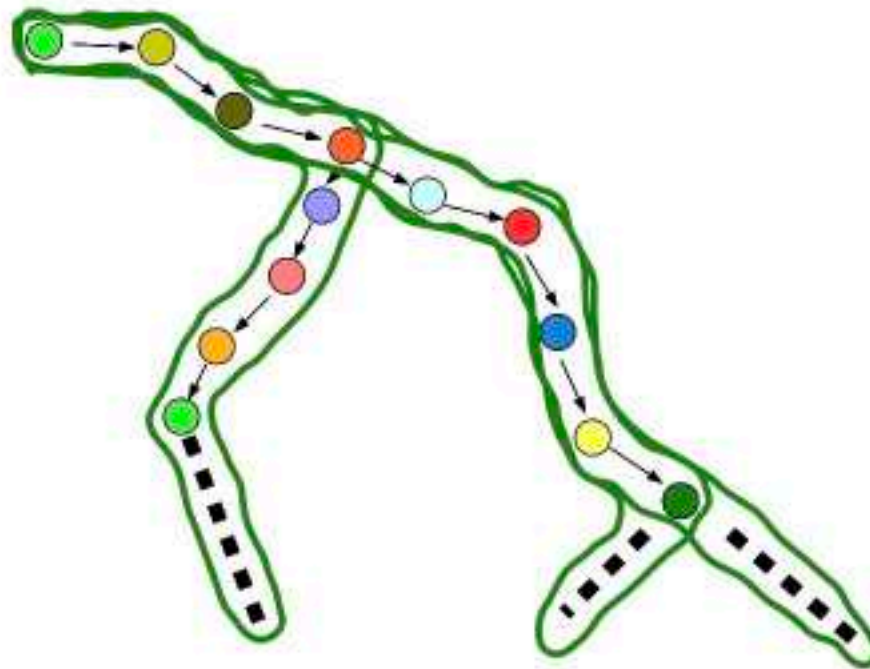
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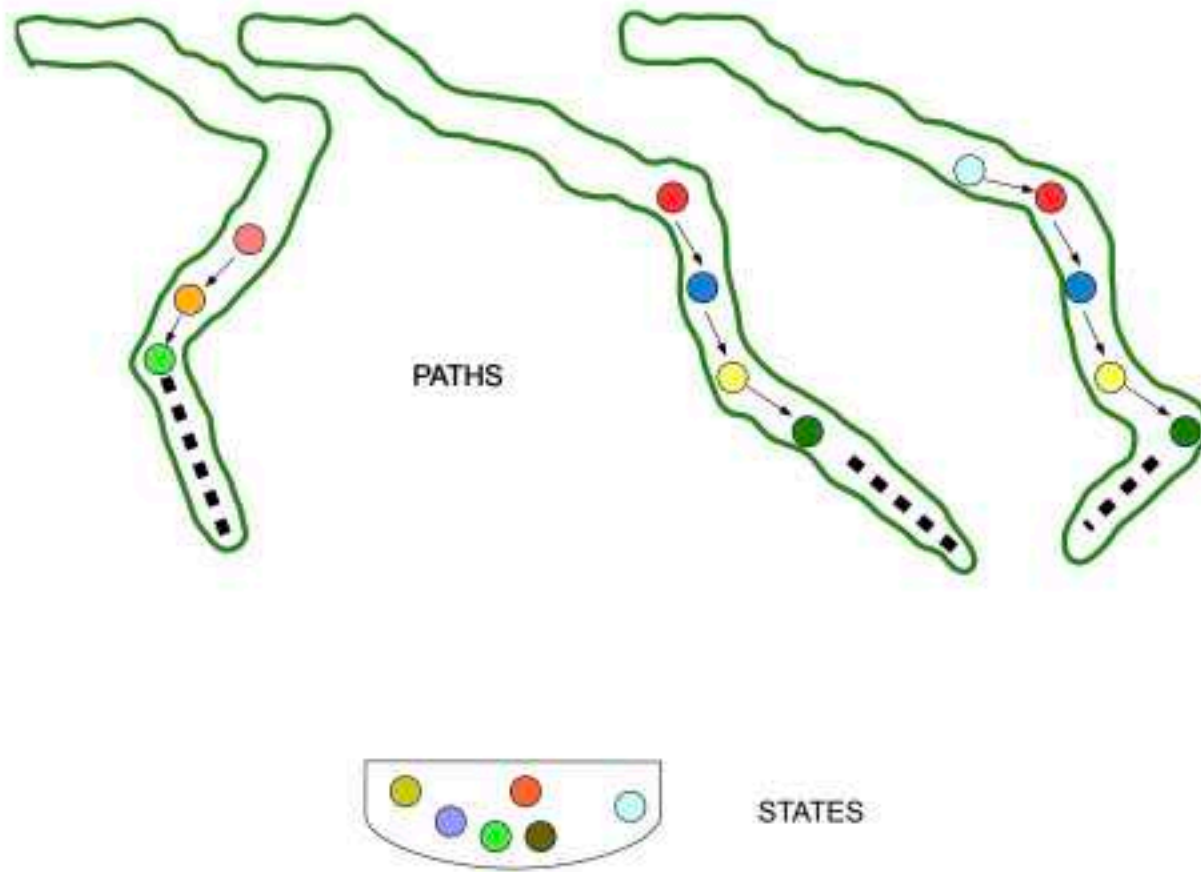
Multiple frontiers for execution

Compute all the distinct paths of system evolution frontiers:
(NOT ALWAYS POSSIBLE)



Collect the states (Reachable States)

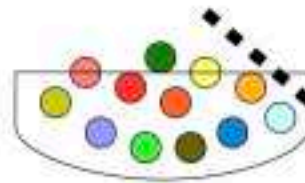
Compute the states through which the system transits: (NOT ALWAYS POSSIBLE!!!!)



Collect the states (Reachable States)

Compute the states through which the system transits: (NOT ALWAYS POSSIBLE!!!!)

PATHS



STATES

Formal Methods

If the trade is so simple, what is the challenge!!!!

- Systems are continuous: infinite states and
- non-terminating: infinite runs
- requirements: temporal

Undecidability with decision procedures!!!!